



MCP sandbox proxy

A separate-origin HTML bridge for the MCP Apps double-iframe boundary.

NestJS proxy page

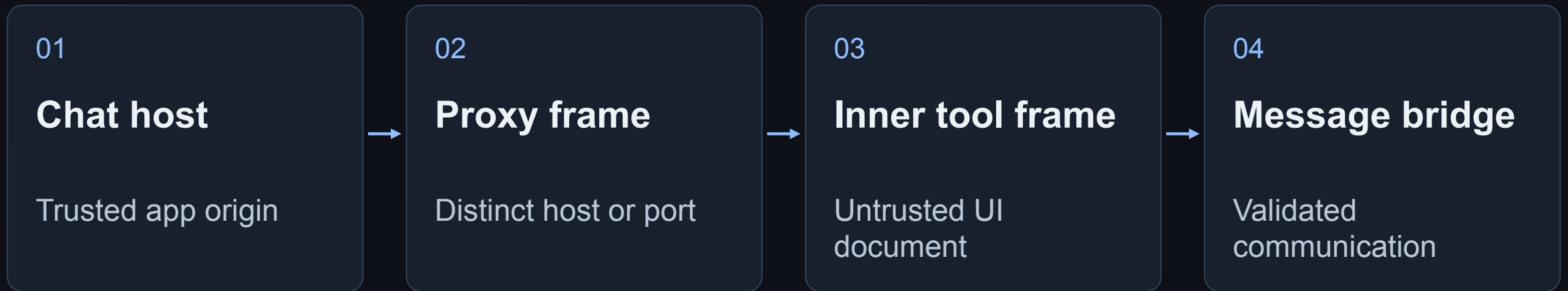
Port 3100

Allowlist required

`apps/mcp-app-sandbox`



Why a separate application exists

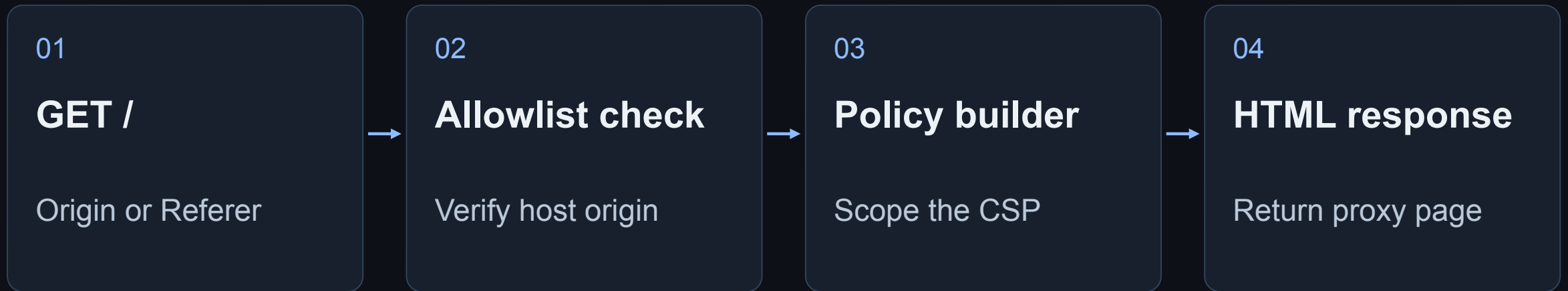


Same-origin hosting would defeat the intended isolation boundary.



Request validation precedes HTML delivery

Allowed request path · invalid origin returns 403 before HTML rendering



No allowlist means no successful page response.



Configure two sides of the connection

The sandbox origin must differ from the chat origin.

```
# Start the sandbox:
MCP_APP_SANDBOX_ALLOWED_HOST_ORIGINS=http://localhost:4207 \
  npm exec nx serve mcp-app-sandbox

# Configure chat-api separately:
MCP_APP_SANDBOX_URL=http://localhost:3100
```



The page and policy are app-owned

01

Controller

Validates the caller context and sets response headers.

02

Service

Combines allowed origin, HTML and CSP construction.

03

Page builder

Creates the bridge document used by the nested renderer.



A rendered MCP view needs the whole chain



Use the `mcp-apps` and `attachment-canvas` decks for the frontend contracts.



Build and check what exists

The resolved project has build, serve, lint and typecheck; no test target.

```
npm exec nx build mcp-app-sandbox
npm exec nx run mcp-app-sandbox:typecheck
npm exec nx run mcp-app-sandbox:lint

docker build -f apps/mcp-app-sandbox/Dockerfile \
  -t mcp-app-sandbox .
```



Limits to carry into operations

01

Exact origin matters

A host allowlist must match the deployed scheme, hostname and port.

02

Not a content service

The app serves a proxy document, not Core storage or user files.

03

Validation gap

No dedicated source tests were discovered in this project.